

ITA0606 – Machine Learning

Exp-15

Program: Perceptron Based IRIS Classification

Program to implement Naïve Bayes Algorithm with Iris Flower Classification

Step 1: Import required libraries

```
from sklearn.datasets import load_iris
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.naive_bayes import GaussianNB
```

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

Step 2: Load the Iris dataset

```
iris = load_iris()
```

Input features

```
X = iris.data
```

Target classes

```
y = iris.target
```

Step 3: Display dataset information

```
print("Feature Names:")
```

```
print(iris.feature_names)
```

```
print("\nTarget Names:")
```

```
print(iris.target_names)
```

Step 4: Split the dataset into training and testing data

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)
```

```
print("\nTraining samples:", len(X_train))
```

```
print("Testing samples:", len(X_test))
```

Step 5: Create Naïve Bayes classifier

```
model = GaussianNB()
```

Step 6: Train the model

```
model.fit(X_train, y_train)
```

Step 7: Predict the test data

```
y_pred = model.predict(X_test)
```

Step 8: Display predicted and actual values

```
print("\nPredicted Values:")
```

```
print(y_pred)
```

```
print("\nActual Values:")
```

```
print(y_test)
```

Step 9: Calculate accuracy

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("\nAccuracy:", accuracy)
```

```
# Step 10: Display confusion matrix
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Step 11: Display classification report
print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=iris.target_names
))
```

OUTPUT

```
Feature Names:
['sepal length (cm)' 'sepal width (cm)'
 'petal length (cm)' 'petal width (cm)']

Target Names:
['setosa' 'versicolor' 'virginica']

Training samples: 120
Testing samples: 30

Predicted Values:
[1 0 2 1 1 0 1 2 1 0 ...]

Actual Values:
[1 0 2 1 1 0 1 2 1 0 ...]

Accuracy: 1.0
```